

## 2018-Ph-H1-Q2

### Section: Our Dynamic Universe

#### Topic: Forces, Energy and Power

A vehicle of mass 1000 kg is travelling at 25 m/s. The brakes apply a constant force and bring it to rest in 5.0 s.

What is the magnitude of the braking force?

#### Solution:

Use Newton's second law. First calculate acceleration:

$$a = \frac{v - u}{t} = \frac{0 - 25}{5.0} = -5.0 \text{ m/s}^2$$

Now apply  $F = ma$ :

$$F = 1000 \times 5.0 = 5000 \text{ N}$$

(Force acts opposite to motion, but the magnitude is **5000 N**.)

**Answer: 5000 N**

#### Revision Tip:

If an object comes to rest, final velocity is zero. Use  $a = \frac{v-u}{t}$  and apply Newton's second law:  $F = ma$ .